

STAT 516: Basic Probability and its Applications

Lecture 4: Discrete and Integer-Valued random variables

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What is a random variable?

- ▶ Often, it is hard and/or impossible to enumerate the entire sample space
- ▶ For a coin flip experiment, the sample space is $\Omega = \{H, T\}$
- ▶ Define a function X s.t. $X(T) = 0$ and $X(H) = 1$
- ▶ X maps the sample space onto the space $\mathcal{D} = \{0, 1\}$

Another motivating example

- ▶ An experiment ξ - rolling a fair die twice
- ▶ The sample space has 36 sample points:

$$\Omega = \{(1, 1), (1, 2), (1, 3), \dots, (6, 4), (6, 5), (6, 6)\}$$

- ▶ Let X be the sum of the two rolls; if $\omega = (1, 1)$ $X = 2$; if $\omega = (6, 5)$, then $X = 11$.
- ▶ Key fact: X is a function of ω – - which is to say, a **random variable**

Formal definition

- ▶ Let Ω be a sample space corresponding to some experiment ξ . A function $X : \Omega \rightarrow \mathcal{R}$ is called a random variable
- ▶ A random variable is **discrete** if its values belong to a countable set

Discrete random variable example

- ▶ The quality control process: we sample batteries (or any other industrially manufactured product) as it comes off the conveyor line. Let F denote the faulty and S the good one. The sample space is $\Omega = \{S, FS, FFS, \dots\}$. Let X be the number of batteries that is examined before the experiment stops. The, $X(S) = 1, X(FS) = 2, \dots$ *infinite but countable outcomes*

Another discrete random variable example

- ▶ Let ξ be the experiment of tossing a coin twice and X the number of times in the two tosses that the head is obtained
- ▶ The four sample points are $HH = \omega_1$, $HT = \omega_2$, $TH = \omega_3$, $TT = \omega_4$.
- ▶ Then, $X(\omega_1) = 2$, $X(\omega_2) = X(\omega_3) = 1$, $X(\omega_4) = 0$.

Probability mass function

- ▶ Let $X : \Omega \rightarrow \mathcal{R}$ be a discrete random variable that takes finite or infinite number of values $x_1, x_2, x_3, \dots$
- ▶ The probability distribution or the **probability mass function** (pmf) of X is the function $p(x) = P(X = x)$ for any $x = x_1, x_2, x_3 \dots$ and 0 otherwise.
 (Handwritten notes: "Big Letter" above X , "random variable" below X , "one of random variable's value." below x)
- ▶ For any pmf, we must have $p(x) \geq 0$ and $\sum_i p(x) = 1$ (Handwritten note: "Probability $0 \leq p \leq 1$ ")

Example: first roll in the game of craps

- ▶ Sample space $\Omega = \{(i, j) : 1 \leq i, j \leq 6\}$ and $P[\{i, j\}] = \frac{1}{36}$
- ▶ The random variable is $X(i, j) = i + j$ with the range $\mathcal{D} = \{2, 3, \dots, 12\}$
- ▶ Easy to put together a pmf of X in the table form; check that e.g. for $P(X = 3) = \frac{1}{18}$

↓
sum=3
 $\begin{pmatrix} (1,2) \\ (2,1) \end{pmatrix} \} \frac{2}{36} = \frac{1}{18}$

$\begin{matrix} (2,6) & (4,4) & & (5,4) & (4,5) \\ (6,2) & (3,5) & & (3,6) & (6,3) \\ & (5,3) & & & \end{matrix}$

x	2	3	4	5	6	7	8	9	10	11	12	...
$P(X)$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$	

center around $x=7$

Cumulative distribution function

- ▶ For a random variable X a **cumulative distribution function** or **cdf** is

$$F(x) = P(X \leq x)$$

for any $x \in \mathcal{R}$
real value of x

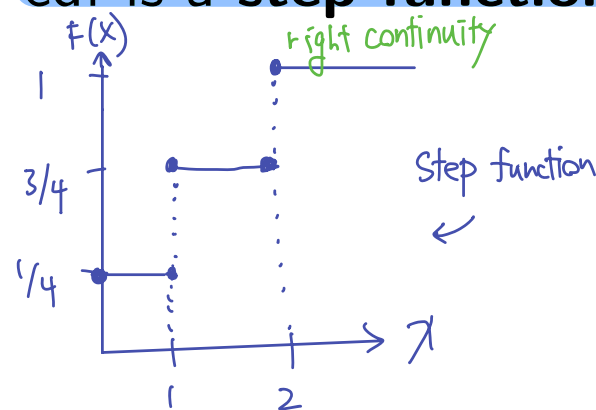
Example I

- ▶ ξ is an experiment of tossing a fair ^{coin}~~die~~ twice; X is the number of heads obtained
- ▶ Possible values of x are $x_1 = 0$, $x_2 = 1$, $x_3 = 2$ ✓
- ▶ $P(X = 0) = P(TT) = \frac{1}{4}$, $P(X = 1) = P(\{HT, TH\}) = \frac{1}{2}$,
 $P(X = 2) = P(HH) = \frac{1}{4}$.
- ✓ ▶ The pmf of X is

x	0	1	2
$p(x)$	0.25	0.5	0.25

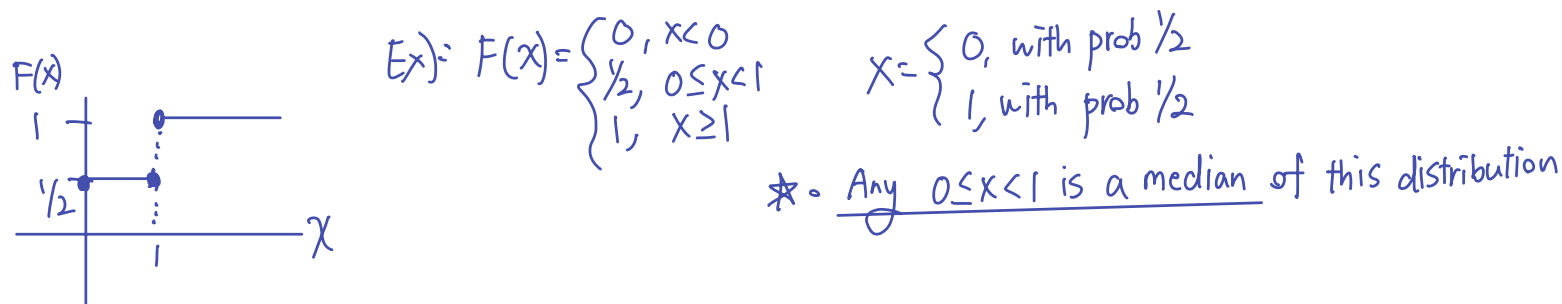
Example I: cumulative density function

- ▶ Clearly, $F(x) = 0$ when $x < 0$
- ✓ ▶ As we reach $x = 0$, a jump occurs: $F(x) = 0.25$ if $0 \leq x < 1$ ^{0.25 only}
- ▶ Next, $F(x) = 0.75$ for $1 \leq x < 2$ ^{0.25+0.5}
- ▶ Finally, $F(x) = 1$ for $x \geq 2$ ^{0.25+0.5+0.25}
- ▶ For discrete random variables a **cdf is a step function**



A median of X

- ▶ Since a cdf of a discrete random variable is a step function, it does not attain all possible values of X
- ▶ How, in general, do we split the distribution into two halves?
- ▶ Any number m such that $P(X \leq m) \geq 0.5$ and also $P(X \geq m) \geq 0.5$ is called a **median** of F (or of X)
- ▶ The median need not be unique

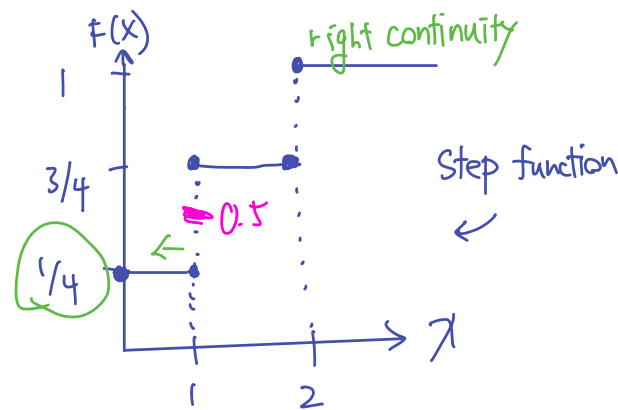


Characterization of a median of X

- ▶ Let X be a random variable with the CDF $F(x)$. Let m_0 be the first number such that $F(m_0) \geq 0.5$ and m_1 the last number such that $P(X \geq x) \geq 0.5$. Then, m is a median of X if and only if $m \in [m_0, m_1]$
- ▶ The proof uses the right continuity of a cdf

Example I): for $m=1$:

- $P(X \leq 1) = 3/4 > 1/2$
- $P(X \geq 1) = 1 - F(1-)$
left
 $= 1 - 1/4 = 3/4 > 1/2$
- $m=1$ is a median



• Median 0.5 is at $x=1$ and in this case it's unique.

Main Properties of cdfs

- A function $F(x)$ is the CDF of some real-valued random variable X if and only if it satisfies all of the following:

1. $0 \leq F(x) \leq 1$ for any $x \in \mathcal{R}$

2. $\lim_{x \rightarrow -\infty} F(x) = 0$

3. $\lim_{x \rightarrow \infty} F(x) = 1$

$$\lim_{x \rightarrow a^+} F(x) = F(a)$$

F(x) converges F(a) from the right as x converges to a from the right

4. Given any real number a , $F(x) \downarrow F(a)$ as $x \downarrow a$

5. Given any two real numbers x, y such that $x < y$,

$F(x) \leq F(y)$. $\left[\begin{array}{l} P(X \leq x) \leq P(X \leq y) \\ \text{whenever } x \leq y \end{array} \right]$

Some remarks

- ▶ Property (4) is called **continuity from the right**. Note that a CDF need not be continuous from the left
- ▶ More precisely, the following is true. Let $F(x)$ be the CDF of some random variable X . Then, for any x ,
 1. $P(X = x) = F(x) - \lim_{y \uparrow x} F(y) = \overset{\text{Right hand}}{F(x)} - \overset{\text{left hand}}{F(x-)}$
 2. $P(X \geq x) = P(X > x) + P(X = x) = (1 - \cancel{F(x)}) + (\cancel{F(x)} - F(x-)) = 1 - F(x-)$

Continuity of probability (Proposition)

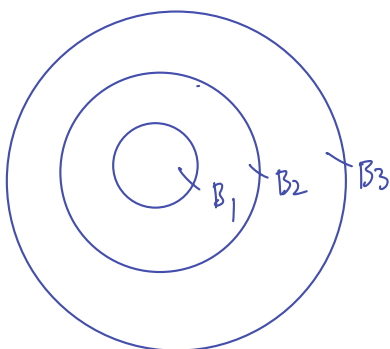
A sequence of events $A_1 \supset A_2 \supset A_3 \supset \dots$

- an infinite family of subsets of S

s.t. $A_n \downarrow A$. $A = \bigcap_{k=1}^{\infty} A_k$

Then, $P(A_n) \rightarrow P(A)$ as $n \rightarrow \infty$

Proof: $B_i = A_i'$, $B_1 \subset B_2 \subset B_3 \subset \dots$
 $\lim_{n \rightarrow \infty} B_n = B$, $B = \bigcup_{i=1}^{\infty} B_i$



$$B_n = \bigcup_{i=1}^n (B_i \setminus B_{i-1}), B_0 = \emptyset$$

• Union of mutually exclusive events

Proof of Property 4.

$$\lim_{n \rightarrow \infty} P(B) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n (B_i \setminus B_{i-1})\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i \setminus B_{i-1}) =$$

$$\sum_{i=1}^{\infty} P(B_i \setminus B_{i-1}) = P(B)$$

$$\text{Since } \bigcup_{i=1}^{\infty} (B_i \setminus B_{i-1}) = B$$

Result: $\lim_{n \rightarrow \infty} P(B_n) = P(B)$